# Portfolio

Jing Yuan

# About me

I'm Jing Yuan, a Secondary 4 student from Singapore who gets excited about understanding how things work and building cool stuff. I'm the kind of person who dives deep into interests. I'll self-teach Fusion 360, pick up competitive programming in C++, and spend hours designing and iterating, just because I'm curious. I like LEGO Technic, I enjoy problem-solving, and I'm driven by that moment when something clicks into place.

# Skill expertise

Programming

- C++
- Python
- Pybricks
- Web Development (React, Tailwind CSS)

CAD & 3D Design

- Fusion 360
- Tinkercad

Electronics

- ESP32-S3
- PID control

Design

- UI/UX Design
- Motion Graphics (Adobe Aftereffects)

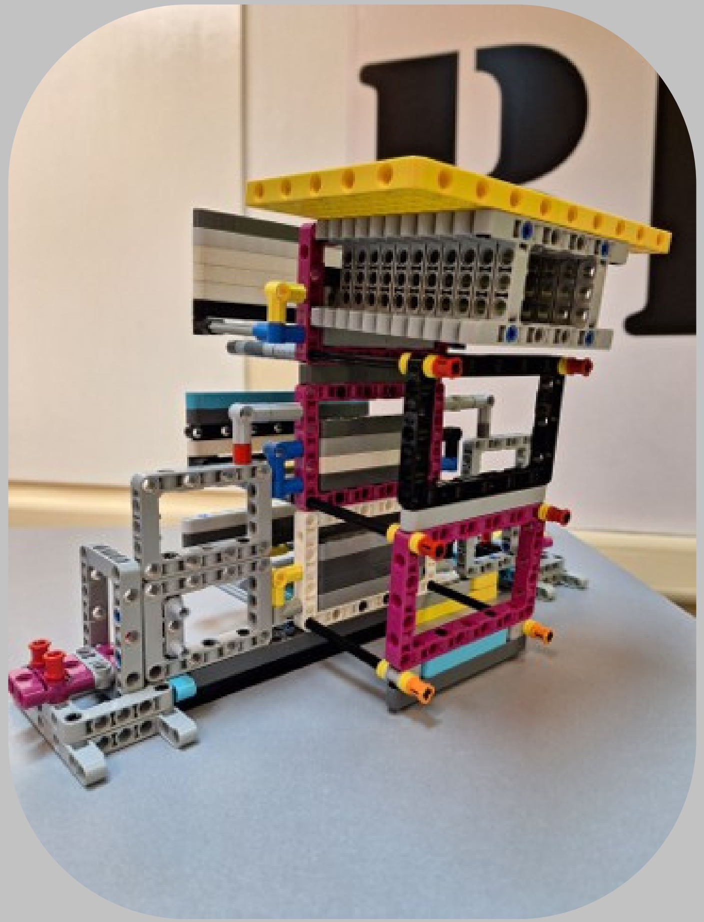
Other

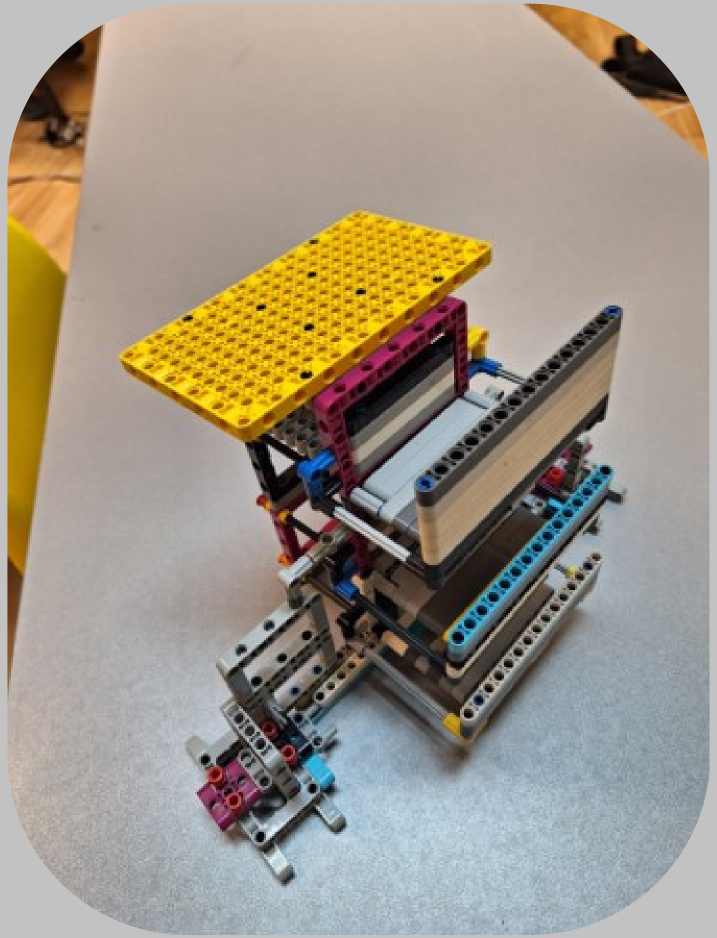
- Competitive Programming (NOI 2026 Finals)
- Game Development(Phaser)

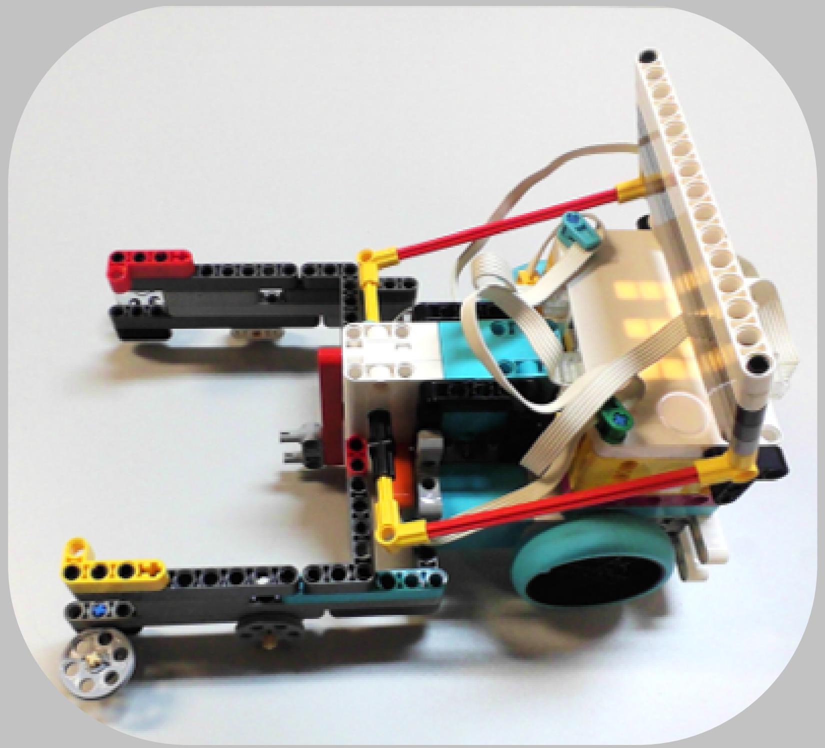
# NRC 2023

## Presentation Finals.

NRC is Singapore's national robotics competition where teams build and program autonomous robots to complete missions on a field
It was my first real robotics competition and my first real taste of what it means to be a builder. The pressure of seeing my own designs on the field, the thrill when they worked, the sting when they didn't. That competition taught me that failure is not the end of world, I just need to learn to improve on it.





# FLL 2023/24

## Finalist

FLL challenges teams to build and program a LEGO robot to complete field missions, alongside a real-world innovation project.
As a new team, we were overambitious, but that taught us to prioritise and plan realistically.
Technically, I developed skills in passive vs active attachments, linkages, and gear mechanics.
Beyond the robot, it was a lesson in teamwork and time management that shaped every season after.

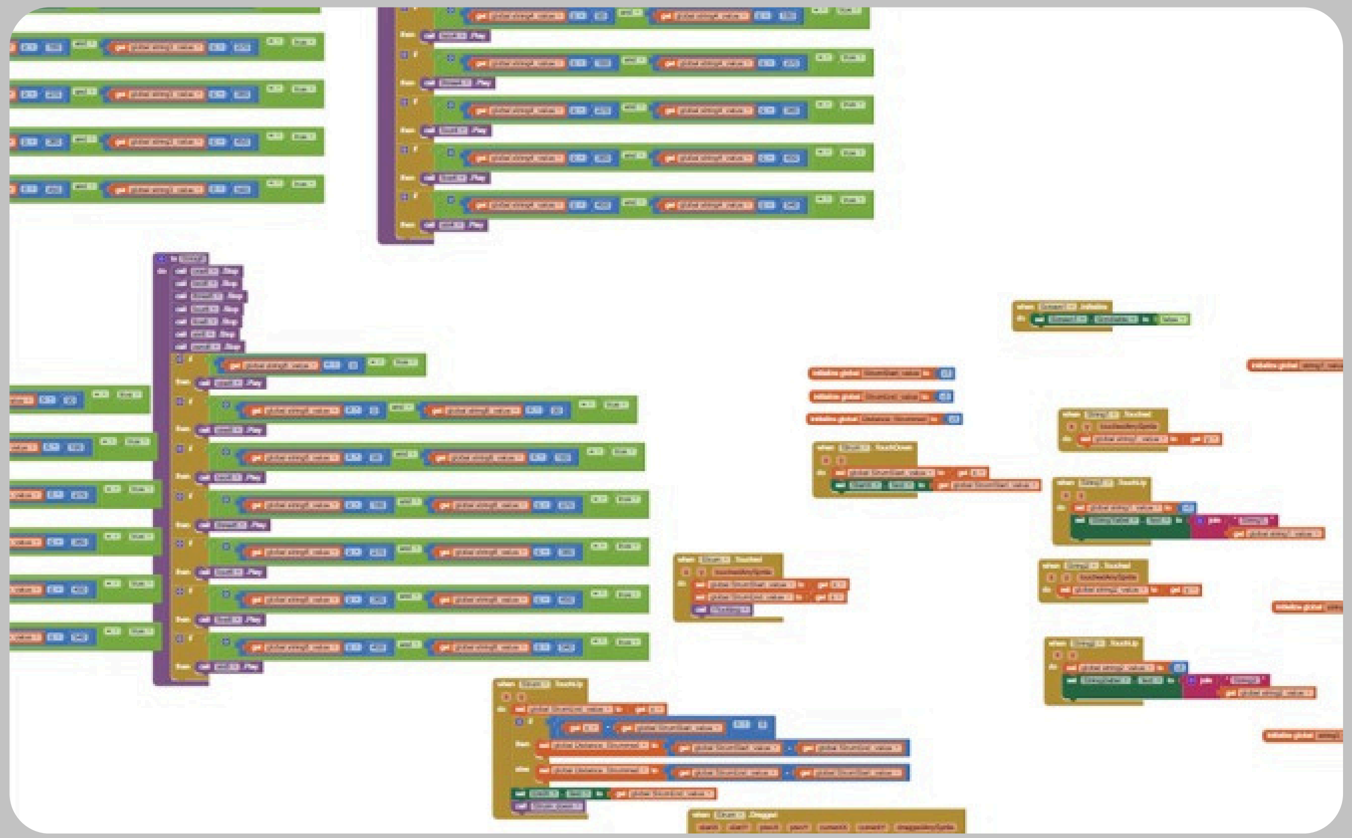
[View all ideal runs]


# Guitar Wizard
# FLL24 Innovation Project

**The Problem:**
Guitar is painful to play, hard to learn, and produces inconsistent sound, making it unappealing to beginners.

**The Solution:**
Guitar Wizard is an app with a virtual fretboard that lets users play guitar without pain or buying an instrument.

**Current Features:**
Virtual fretboard to play chords and tabs with realistic sound frequency adjustments.



**Bulid using MIT app inventor**

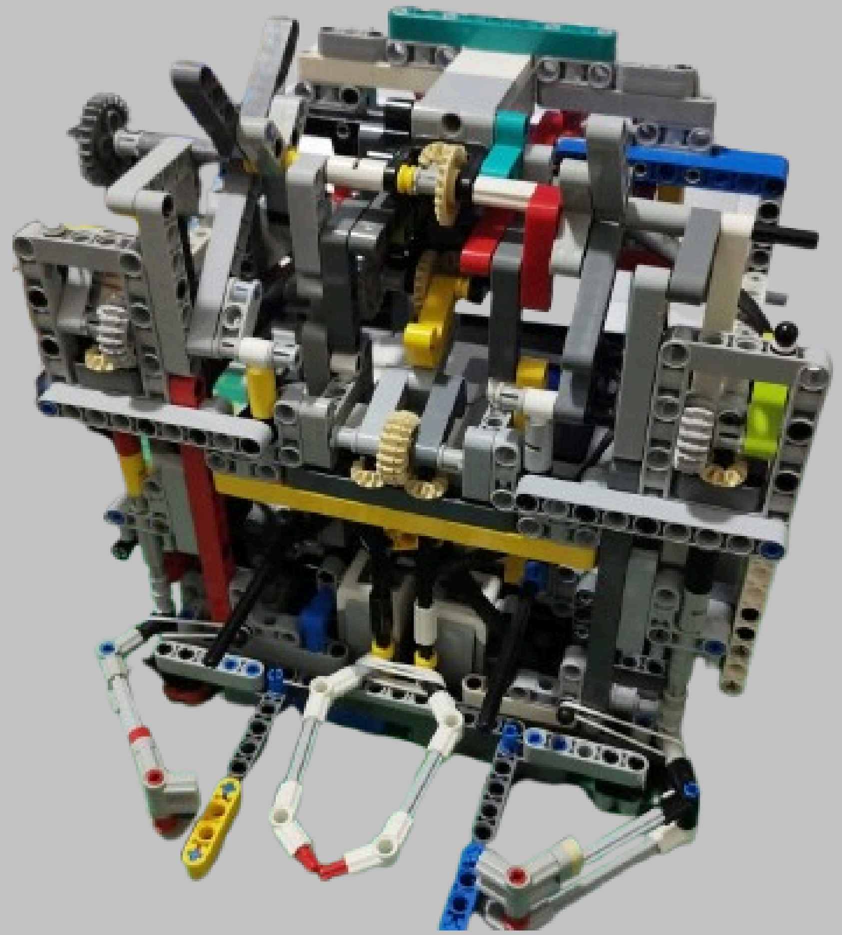
# WRO 2024

## 7th place

This season pushed me deeper into sensor programming. Learning the difference between DLT and SLT functions changed how I thought about robot perception. A key breakthrough was using RGB and HSV values to identify element colours and program logic to decide which elements to collect.

We placed 7th at Nationals, close, but I wanted first next time

[View ideal full run]



# FLL 2024/25

## Championship award, Best Robot Run

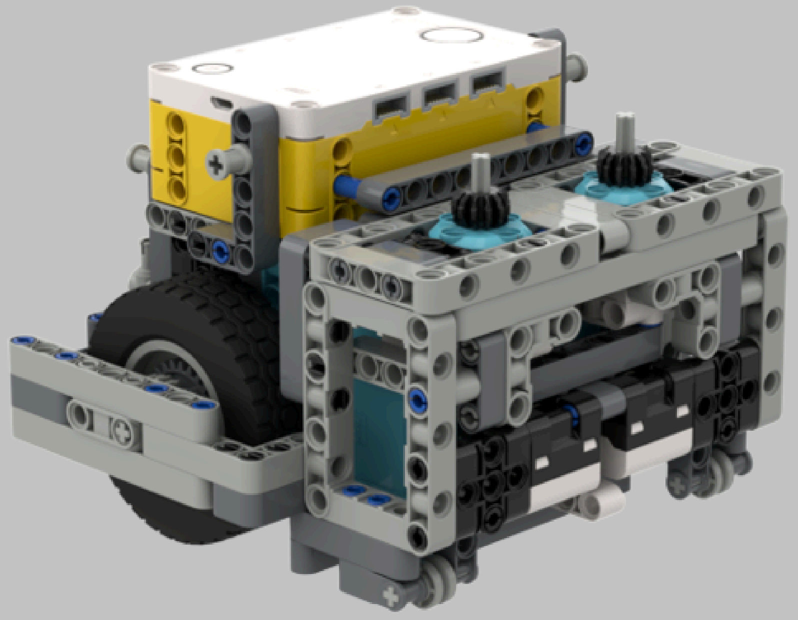
We were the first team in the competition to use PyBricks — and for all of us, it was our first time with text-based programming entirely. That meant rebuilding from scratch: rewriting movement functions like straight, spot turn, SLT and DLT in Python, implementing PID for consistency, and running repeated tests to verify reliability before competition day.

It wasn't easy. But we came out National Champions with a Best Robot Run award. Even then, we didn't hit our target of scoring 585 out of 620, which told us there was still more to chase.





[Learn more about FLL24/25]()



# Innovation Project

**The Problem:**

Divers face three major safety challenges underwater:

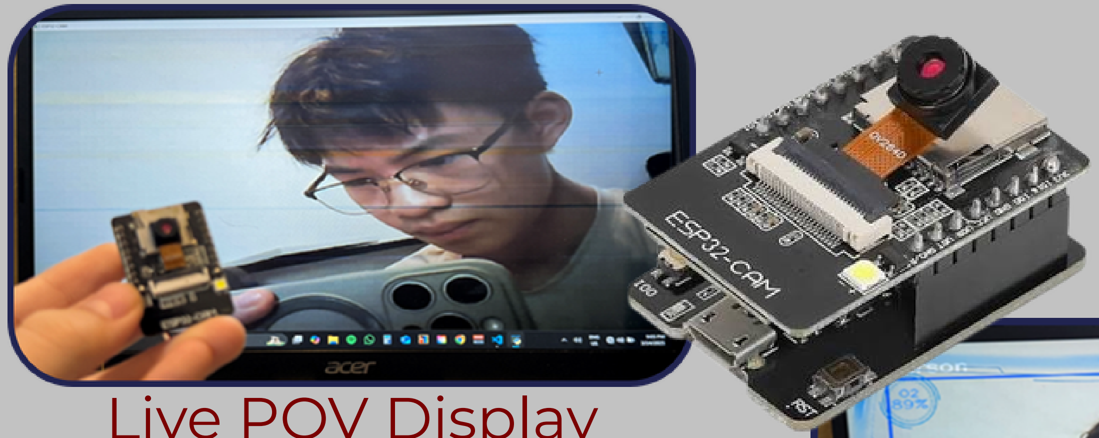
- Hard to monitor other divers' status
- Difficult to keep track of multiple gauges (oxygen, pressure, depth, etc.)
- Risk of colliding with obstacles due to poor visibility

**The Solution:**

SCUBATHON — a heads-up display (HUD) system that helps divers monitor themselves and their team while diving safely.

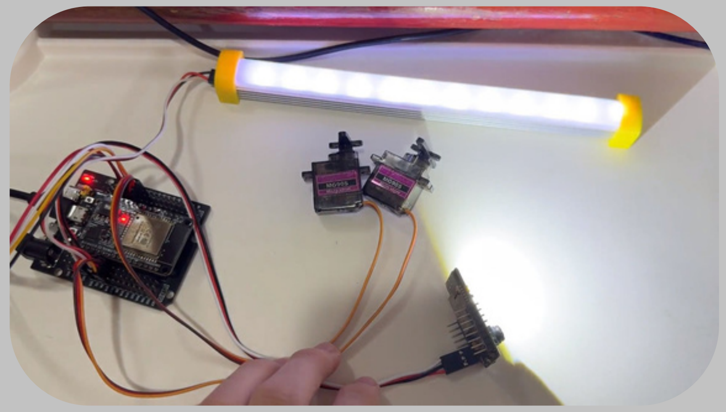
What it does:

1. Live POV Display — Shows the diver's camera feed on a screen
2. HUD Overlay — Displays real-time status information (gauges, vitals) on the video
3. LED Alerts — Uses lights controlled by ESP32 to provide visual warnings
4. Ultrasonic Detection — Alerts divers when obstacles are near
5. Voice Commands — Divers can control the system hands-free



Live POV Display

HUD Overlay


LED Alerts

Ultrasonic Detection

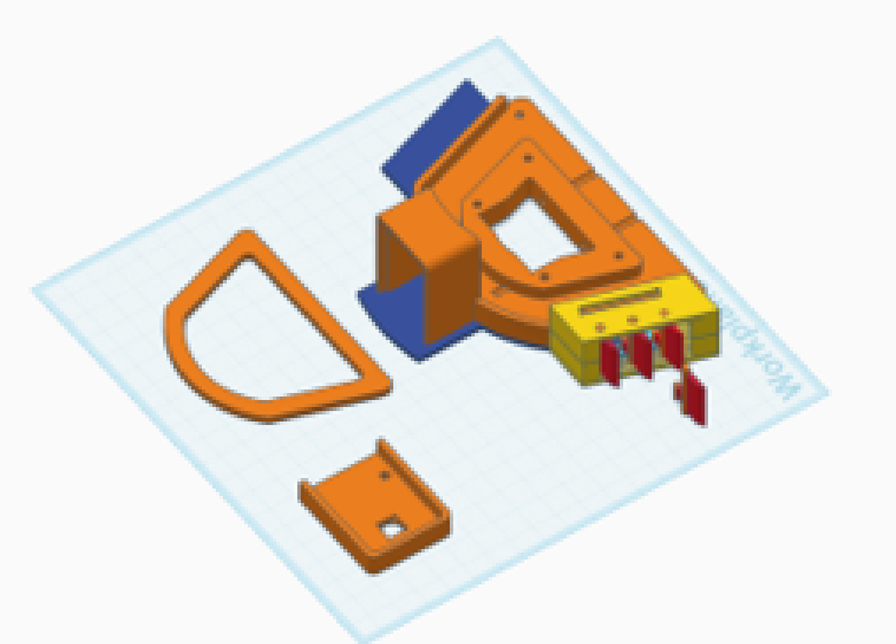
# Innovation Project

Tech Stack:
Languages: Python, C++
Vision: OpenCV + YOLO (for human detection)
Hardware: ESP32 (microcontroller), ultrasonic sensors, servo motors, LED strips
3D Design: TinkerCAD for custom compartments
Development: Arduino IDE, Visual Studio Code

[Learn more about the FLL24/25 Innovation Project]

| | |
|---|---|
| OPENCV | Video Capture |
| YOLO | Human detection |

| | |
|---|---|
| ESP32Servo | Control Servo |
| FastLED | Led effects |
| DFRobot_DF2301Q | Voice commands |
| NewPing | Ultrasonic sensor |



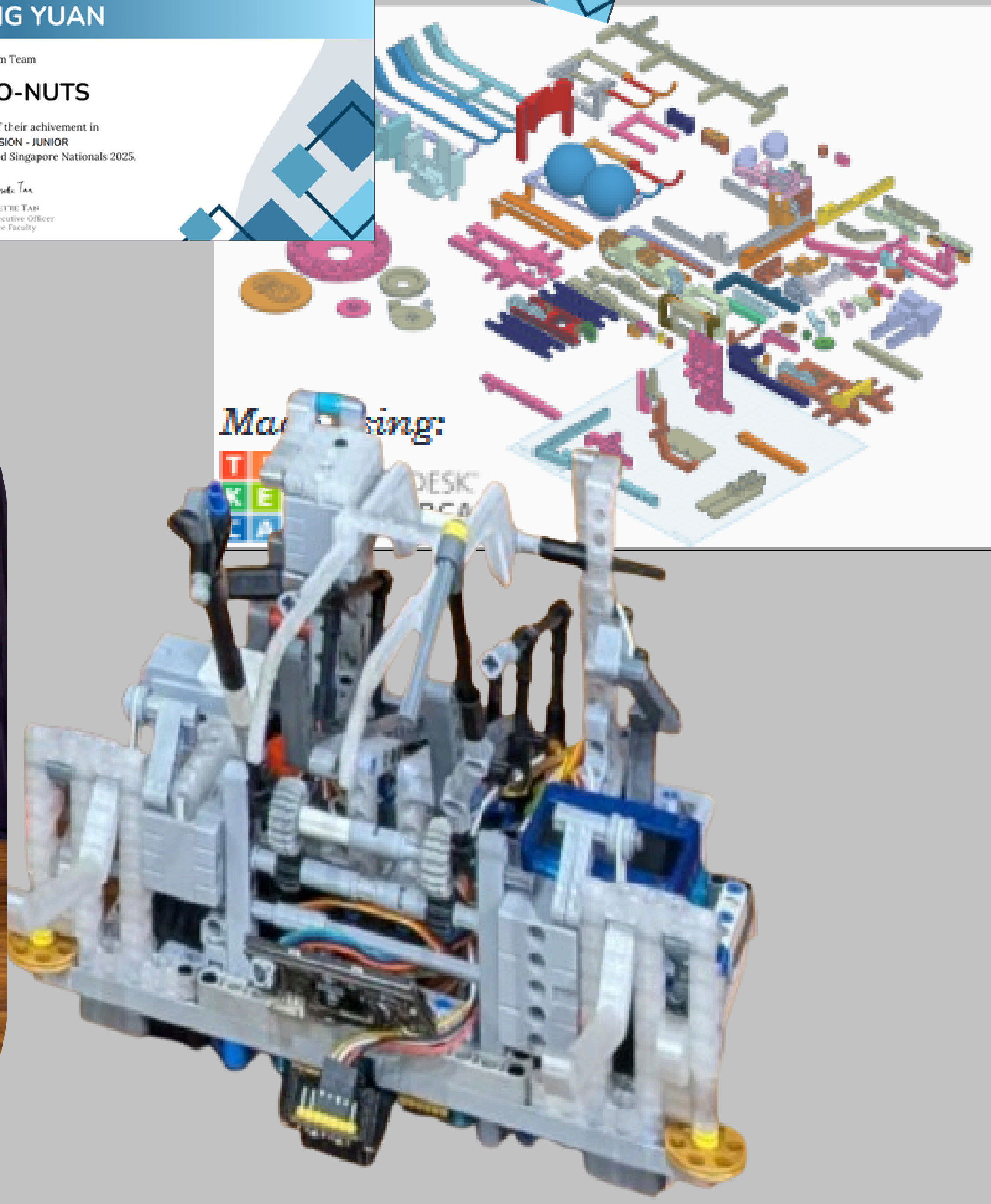
3D Design: TinkerCAD for custom compartments

# WRO 2025

## Championship award, Top Score-(Open Hardware) ,Side Quest 4

The first year WRO allowed open hardware meant learning everything from scratch, C++, sensor libraries, Arduino IDE, 3D printing custom parts in Tinkercad, and rewriting PID movement functions on entirely new hardware. New sensors like the HuskyLens added another layer to figure out.
The steepest learning curve we'd faced. We came out National Champions with Top Score and Side Quest 4 (Fastest time to complete a task).

[Learn more about WRO25]


# WRO 2025 International

## Sliver award

Competing on home ground in Singapore against 112 teams brought a different kind of pressure. We came in more prepared — redesigning our front mechanism from a hook to a full grabber for greater versatility, and improving our robot's overall capability.

But preparation only goes so far. Where we fell short was executing difficult surprise missions under time pressure,  something we hadn't drilled enough. Finishing 35th with a Silver Medal, I couldn't help but feel we had more to give.

**Next time, we're going for 1st.**

[Learn more about WRO25]


TEAM SG

# NOI

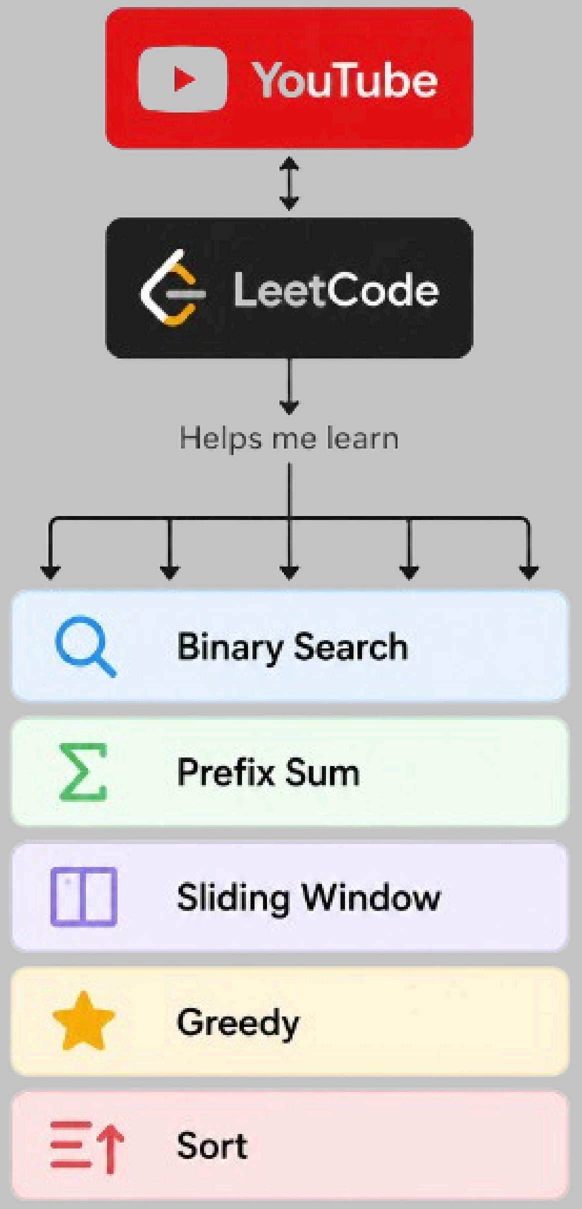
## Finalist

Making it to NOI finals was a challenge I wasn't fully prepared for. Walking in, I knew I was competing against talented programmers. The pressure was real, but it pushed me to give my best.

Every breakthrough made it worth it, figuring out tricky algorithms, realizing new approaches mid-contest, getting solutions accepted after debugging. Those 'aha moments' stayed with me. The hardest part wasn't the algorithms—it was overthinking my own solutions. I'd second-guess correct approaches and try different ones. That's something I'm working on.

I did NOI purely for the challenge, not for medals. If I get another opportunity, I'd definitely compete again.

[Learn more about NOI]




CERTIFICATE OF PARTICIPATION

This certificate is presented to

Lee Jing Yuan

of Admiralty Secondary School
for attending and participating in
The 29th National Olympiad in Informatics
organized by the School of Computing, National University of Singapore
on 14th March 2026

Associate Professor Steven Halim
Chairman
NOI Organising Committee

NUS National University of Singapore
School of Computing

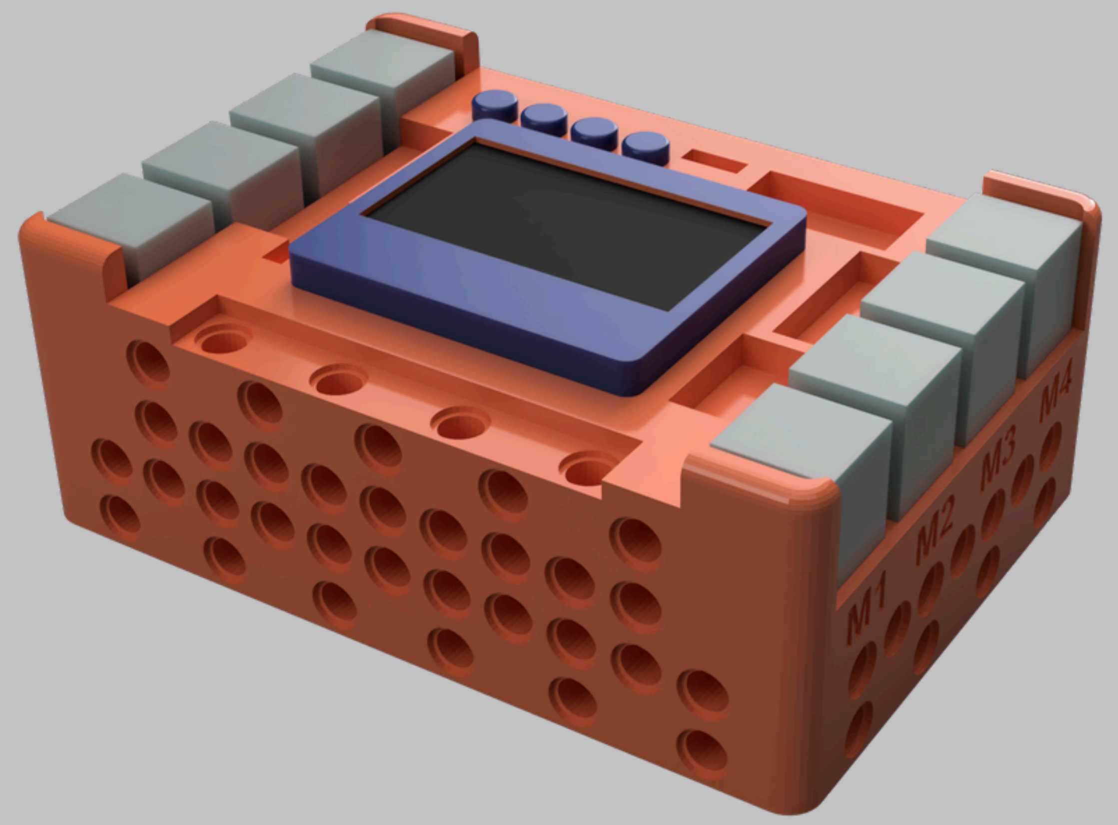
Professor Tulika Mitra
Dean
School of Computing

# EVO

## What is EVO?

EVO is a custom microcontroller built on the ESP32-S3, designed by my CCA coach to overcome LEGO controller limitations. Standard controllers like EV3 and SPIKE Prime cap out at 4 motors and 4 sensors combined. EVO enables teams to connect far more sensors, servos, and motors while remaining WRO legal — taking full advantage of the 2025 rule change allowing open electronics.

[Learn more about EVO]



# EVO

## My Contributions

- PID Movement Functions — Wrote foundational PID functions for straight-line stabilization and spot turns, forming the core movement library for competition robots.
- Servo Library Testing — Evaluated an alternative servo library to resolve compatibility issues with the existing implementation.
- Hardware Stress Testing — Pushed EVO to its limits with simultaneous servos, rapid loops, and precision checks to surface bugs and edge cases.
- Code Review & Bug Reporting — Conducted code reviews and logged bugs to improve reliability.
- EVO Beta Tester Certificate — Received official certification for beta testing and development contributions to the EVO microcontroller prior to its public release.

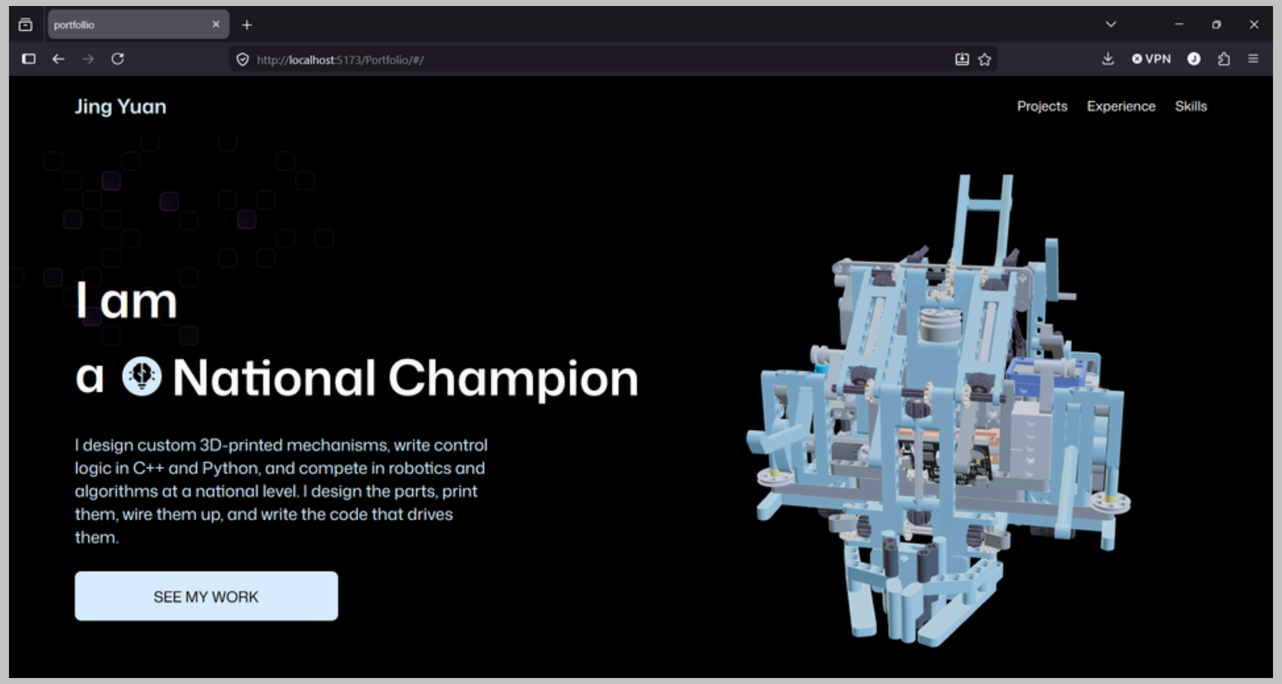
# Portfolio website

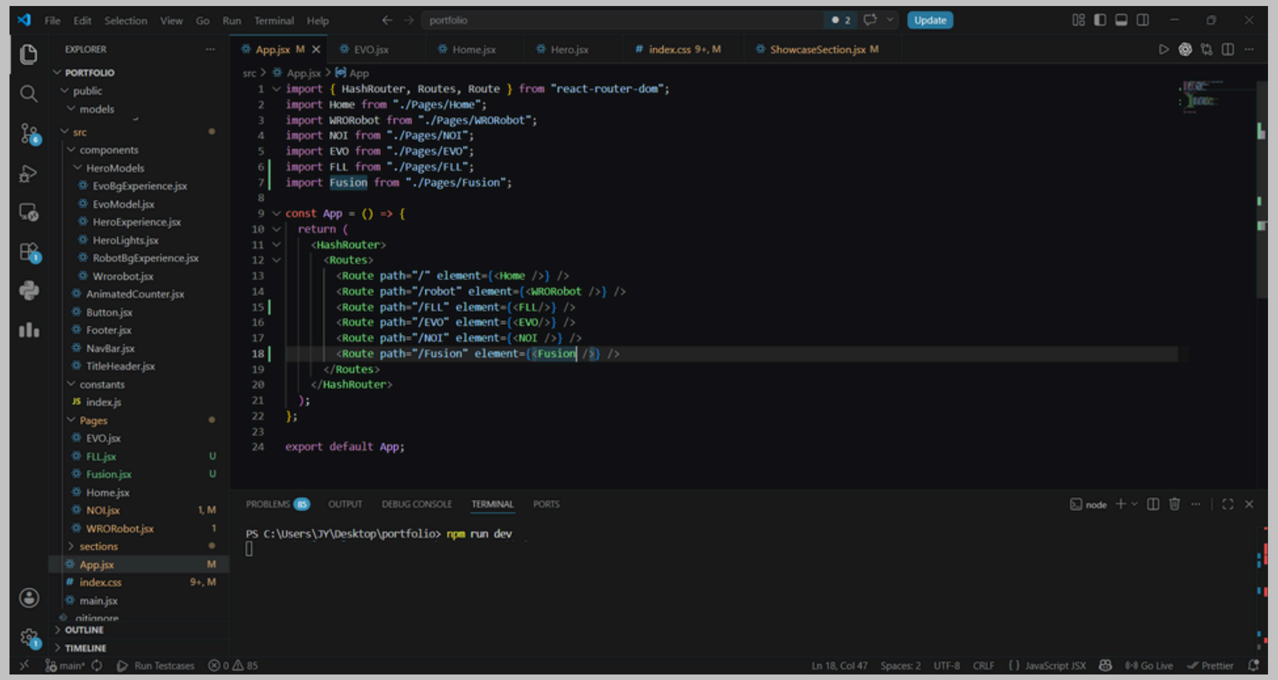
I started this project because I wanted to break into web development but never had the opportunity before. I began with the fundamentals—HTML and CSS—following along with tutorials, but quickly realized I needed more depth.

After searching YouTube, I found a 3D portfolio tutorial that matched my vision. Rather than blindly copying the code, I decided to understand it first. I spent time learning React and JavaScript fundamentals, watching tutorials until I felt confident with the core concepts. Only then did I tackle the 3D portfolio tutorial, which let me customize and modify elements to fit my needs. Whenever I hit a wall, I'd ask AI to explain the concepts I didn't grasp.

This approach—learning foundations before diving into projects—meant I could adapt the code rather than just replicate it, turning someone else's tutorial into something truly mine.

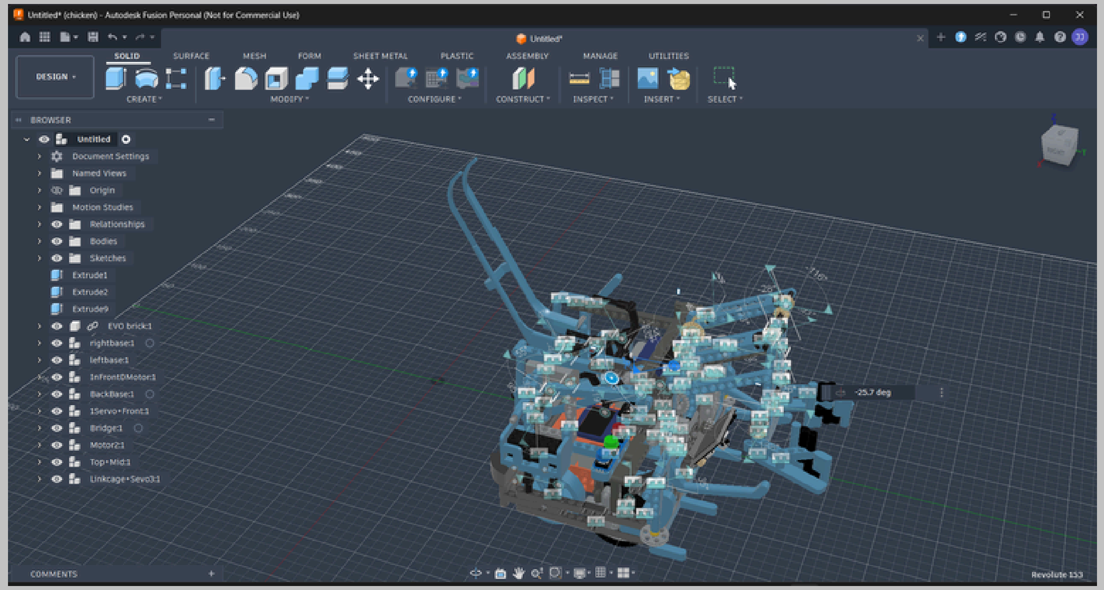
[View my website]



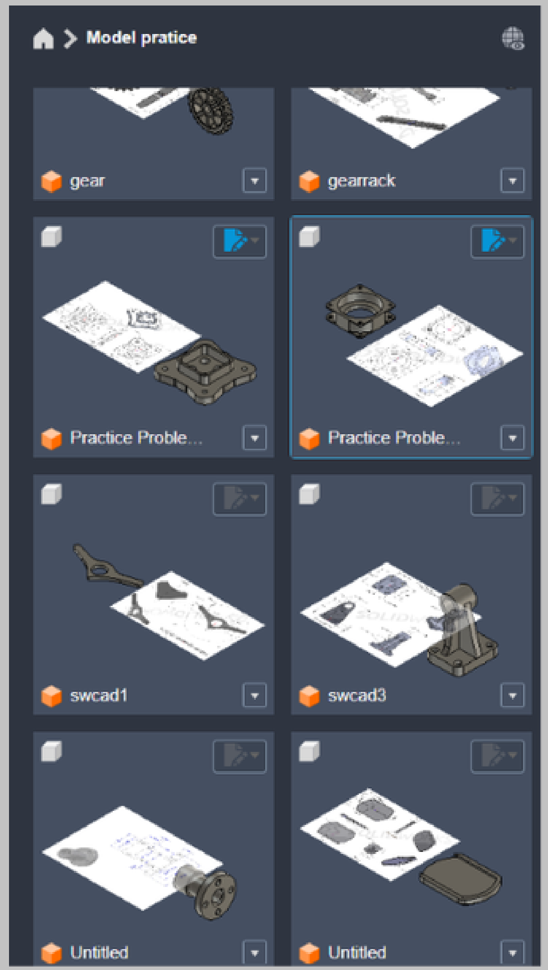



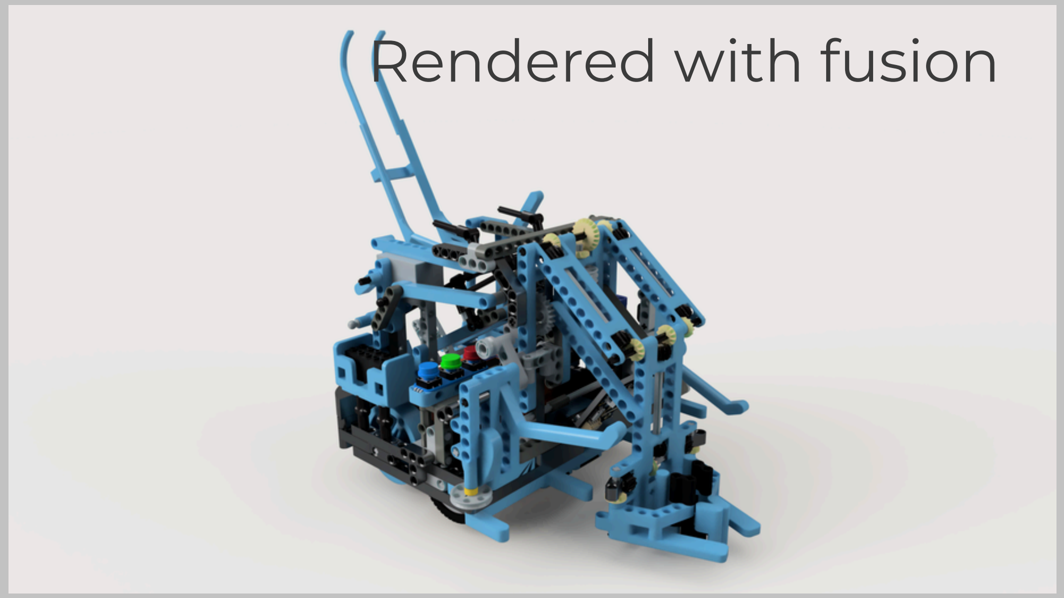
# Fusion 360

My WRO robot took months of design and building. I wanted to keep a memory of it, so I decided to recreate it in Fusion 360. Stepping up from Tinkercad to professional CAD, I hunted down STL models for every component, converted them to STEP files, and spent a month reconstructing it virtually piece by piece. Adding working joints to animate the mechanisms gave the model real functionality. I also modeled random CAD drawings I found online, just learning by doing. Through this work, I've levelled up my CAD skills significantly.

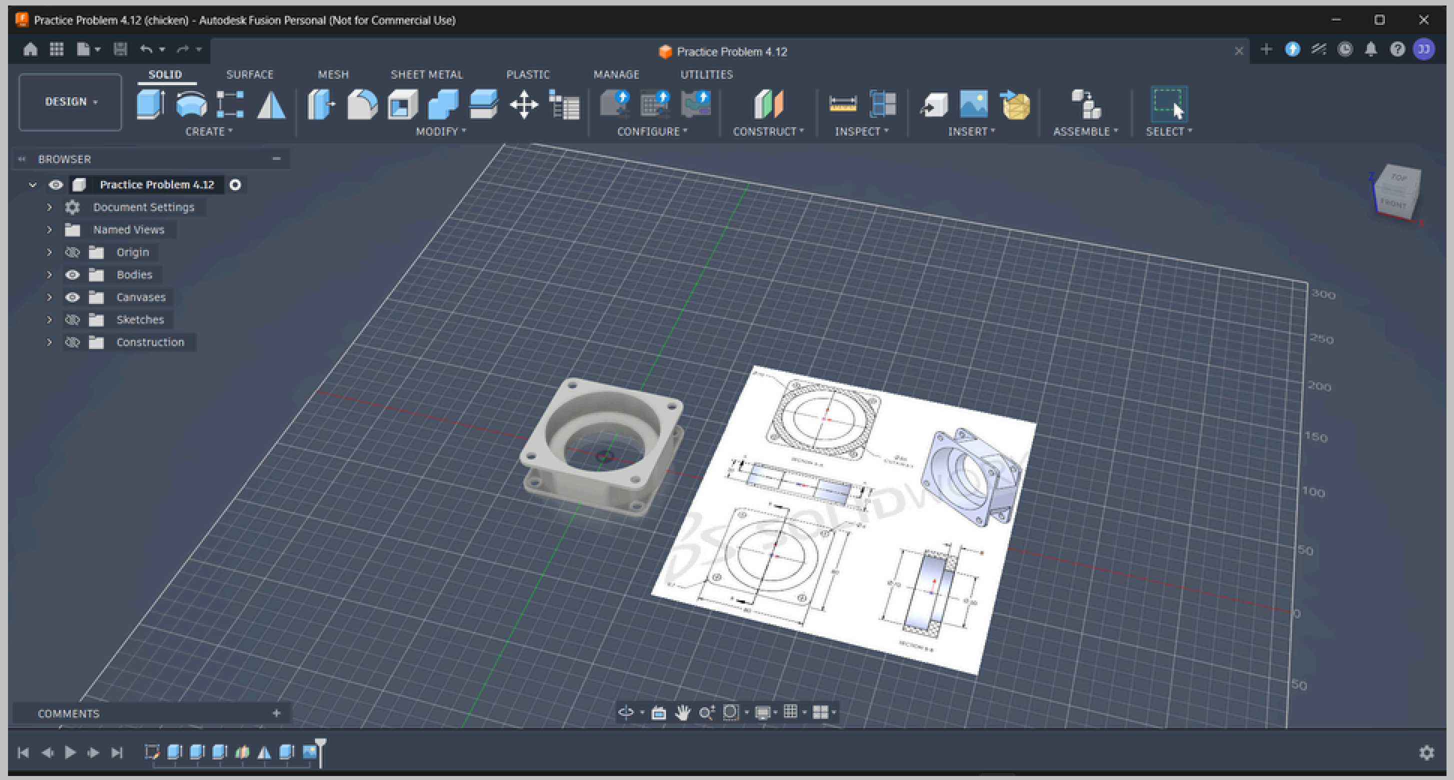


Over 80 different componets



Rendered with fusion



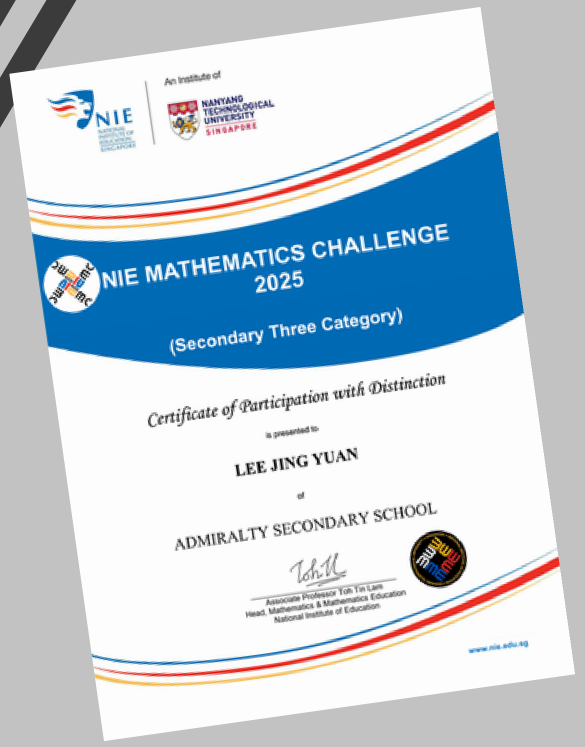
# Academic Achievements

## SCHOLARSHIPS & AWARDS

- Edusave Certificate of Academic Achievement - 2024
- Edusave Scholarship - 2025
- EAGLES Award (Achievement, Good Leadership and Service) - 2025
- Semestral Exemplary Admiral Award - 2026 (Awarded to 5 students in the class)


## MATHEMATICS & SCIENCE OLYMPIADS

- Singapore and Asian Schools Math Olympiad (SASMO) - Silver Award - 2024
- NIE Mathematics Challenge - 2025 - Participation with Distinction
- Singapore Junior Physics Olympiad (SJPO) - Certificate of Participation - 2025
- Singapore and Asian Schools Math Olympiad (SASMO) - Certificate of Participation - 2026
- Singapore Junior Chemistry Olympiad (SJCHO) - Certificate of Participation - 2026
- Singapore Junior Physics Olympiad (SJPO) - Certificate of Participation - 2026
- Singapore Physics League (SPhL) - Certificate of Participation - 2026




# Leadership & Impact

## LEADERSHIP & SERVICE


- Chinese Language Representative - 2023
- ICT Representative - 2024
- Robotics Club Executive Committee (2025-2026) — Managed equipment distribution and developed PyBricks template for junior members
- Class Vice Chairperson - 2026 — Coordinated class logistics and events

## COMMUNITY ENGAGEMENT & GROWTH

- Overseas Cultural Immersion Learning Journey to Penang - 2024
- Sunlove Active Aging Centre - 2025 (Engaged elderly residents through terrariums and games)
- Coastal Cleanup at Coney Island - 2026 (Self-registered environmental conservation initiative)
- Admirals Run - 2026 (10th place in cohort through consistent training)


# Skills & Development

## UI/UX & GAME DEVELOPMENT

- UI/UX Figma course - 2025
- Phaser Game course - 2026

## APPLIED LEARNING EXPERIENCES

- APLM@SP Awesome Motion Graphics - 2025
- APLM@RP The Amazing Wonders of Chemistry - 2025

SINGAPORE POLYTECHNIC

*Certificate of Completion*

This is to certify that

**LEE JING YUAN**

has successfully completed the Applied Learning Module (ApLM)

**Awesome Motion Graphics**

Module Code : ApLMSPMD007
Grade : Pass with Commendation
School : Admiralty Secondary School
Date : 03-Jun-2025 to 05-Jun-2025

Loh Yew Chiong
Deputy Principal
Singapore Polytechnic

SP007947D2222-1

REPUBLIC POLYTECHNIC

*Certificate of Completion*

This is to certify that

**LEE JING YUAN**

has successfully completed the Applied Learning Module (ApLM)

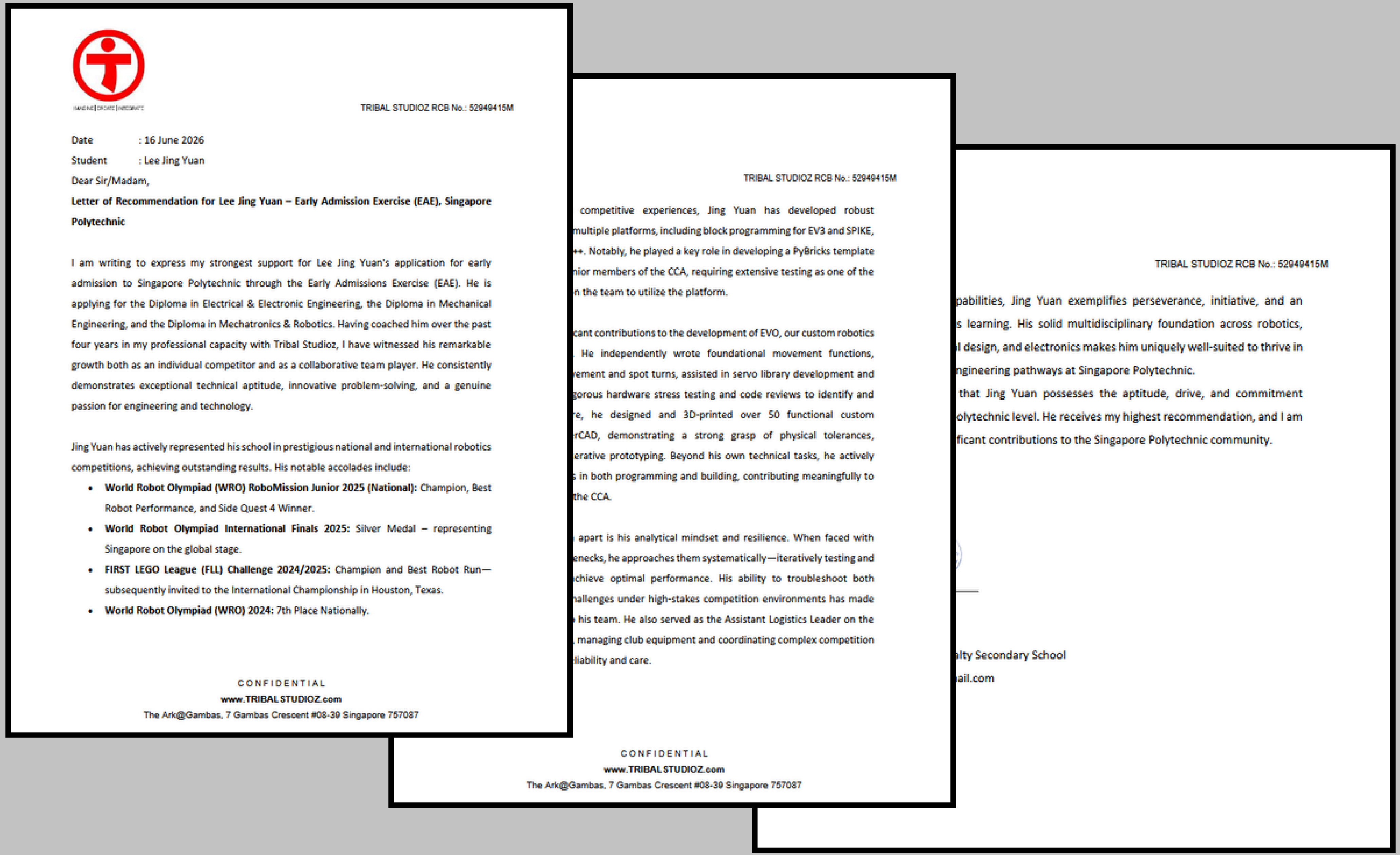
**The Amazing Wonders of Chemistry**

Module Code : ApLMRPMS002
Grade : Pass with Commendation
School : Admiralty Secondary School
Date : 27-Oct-2025 to 29-Oct-2025

Boo Chong-Han
Deputy Principal (Academic & Organisational Development)
Republic Polytechnic

RP002947D2766-1

# Testimonial



TRIBAL STUDIOZ RCB No.: 52949415M

Date : 16 June 2026

Student : Lee Jing Yuan

Dear Sir/Madam,

**Letter of Recommendation for Lee Jing Yuan – Early Admission Exercise (EAE), Singapore Polytechnic**

I am writing to express my strongest support for Lee Jing Yuan's application for early admission to Singapore Polytechnic through the Early Admissions Exercise (EAE). He is applying for the Diploma in Electrical & Electronic Engineering, the Diploma in Mechanical Engineering, and the Diploma in Mechatronics & Robotics. Having coached him over the past four years in my professional capacity with Tribal Studioz, I have witnessed his remarkable growth both as an individual competitor and as a collaborative team player. He consistently demonstrates exceptional technical aptitude, innovative problem-solving, and a genuine passion for engineering and technology.

Jing Yuan has actively represented his school in prestigious national and international robotics competitions, achieving outstanding results. His notable accolades include:

- **World Robot Olympiad (WRO) RoboMission Junior 2025 (National):** Champion, Best Robot Performance, and Side Quest 4 Winner.
- **World Robot Olympiad International Finals 2025:** Silver Medal – representing Singapore on the global stage.
- **FIRST LEGO League (FLL) Challenge 2024/2025:** Champion and Best Robot Run—subsequently invited to the International Championship in Houston, Texas.
- **World Robot Olympiad (WRO) 2024:** 7th Place Nationally.

CONFIDENTIAL
www.TRIBALSTUDIOZ.com
The Ark@Gambas, 7 Gambas Crescent #08-39 Singapore 757087

TRIBAL STUDIOZ RCB No.: 52949415M

competitive experiences, Jing Yuan has developed robust
multiple platforms, including block programming for EV3 and SPIKE,
++. Notably, he played a key role in developing a PyBricks template
nior members of the CCA, requiring extensive testing as one of the
n the team to utilize the platform.

cant contributions to the development of EVO, our custom robotics
. He independently wrote foundational movement functions,
ement and spot turns, assisted in servo library development and
gorous hardware stress testing and code reviews to identify and
e, he designed and 3D-printed over 50 functional custom
rCAD, demonstrating a strong grasp of physical tolerances,
erative prototyping. Beyond his own technical tasks, he actively
s in both programming and building, contributing meaningfully to
the CCA.

apart is his analytical mindset and resilience. When faced with
enecks, he approaches them systematically—iteratively testing and
chieve optimal performance. His ability to troubleshoot both
hallenges under high-stakes competition environments has made
his team. He also served as the Assistant Logistics Leader on the
managing club equipment and coordinating complex competition
liability and care.

CONFIDENTIAL
www.TRIBALSTUDIOZ.com
The Ark@Gambas, 7 Gambas Crescent #08-39 Singapore 757087

TRIBAL STUDIOZ RCB No.: 52949415M

pabilities, Jing Yuan exemplifies perseverance, initiative, and an
s learning. His solid multidisciplinary foundation across robotics,
l design, and electronics makes him uniquely well-suited to thrive in
ngineering pathways at Singapore Polytechnic.

that Jing Yuan possesses the aptitude, drive, and commitment
olytechnic level. He receives my highest recommendation, and I am
ficant contributions to the Singapore Polytechnic community.

alty Secondary School
ail.com

[View Full Testmonial]

# Thank You